WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

Technical Specification for AC 3 Phase 4 Wire Solid State (Static) Import / Net (Import - Export) Smart Energy Meter of voltage levels 132KV,33 KV,11KV and LT CT operated 3x240V

1.0 SCOPE

This scope covers design, engineering, manufacture, testing, inspection and supply of AC 3 Phase 4 Wire Solid State (Static) Import /Net (Import/Net (Import-Export) Smart Energy Meter with backlit LCD display used for balanced/unbalanced load. The Smart Meter shall be capable of recording and displaying energy in **MWh /KWh** & demand in **MVA / KVA** as the case may be, in all four quadrants. Smart Meter shall have facility / capability of recording tamper information & load survey of active energy (both Import/Net (Import - Export)), apparent energy, reactive energy, phase currents, phase voltages & other parameters in non volatile memory.

It is not the intent to specify completely herein all the details of the design and construction of Smart Meter. However the Smart Meter shall conform in all respect to high standards of engineering, design and workmanship and shall be capable of performing commercial operation continuously in a manner acceptable to WBSEDCL, who will interpret the meanings of drawings &specification and shall have the right to reject any work or material which in its judgment is not in accordance herewith. The Smart Meter shall be complete with all components, accessories necessary for their effective and trouble free operation for the purpose mentioned above. Such components shall be deemed to be within the scope of bidder's supply irrespective of whether those are specifically mentioned or not in this specification or in the commercial order.

2.0 STANDARDS APPLICABLE

Unless specified elsewhere in this specification, the performance & testing of the Smart Meters shall conform to the following Indian / International standards, to be read with up-to-date and latest amendments / revisions there of as on 90 days prior to floating of tender.

Sl.No.	Standard No.	Title
1	IS 14697 (1999)	AC Static Transformer Operated Watt hour and VA -hour Meters
2	**IEC 62053-22 with latest amendments**	Electricity Metering Equipment (A.C.) - Particular Requirements - Part 22: Static Meters for Active Energy (Classes 0.2S and 0.5S)
3	CBIP Report No. 325 along with its latest amendments, if any	Specification for AC Static Electrical Energy Meters
4	IS 16444 Part II (Smart Features)	Specification for AC Static Transformer Operated Watt hour And Var-Hour **Smart** Meters, Class 0.2s, 0.5s
5	IS 11731 (Part-I)	Specification of material of terminal block.
6	ISO 75-1&2 (1993)	Specification of temperature test of materials.
7	IS 12346 :1988	Specification for testing equipment for AC Static Electrical Energy Meter (latest amendment)
8	**IS 9000**	Basic Environmental Testing Procedures for Electronic & electrical Items
9	IS-15959: (DLMS) with latest amendment	Data Exchange for Electricity Meter Reading Tariff & Load Control - Companion Specification

3.0 CLIMATIC CONDITIONS

The Smart Meters to be supplied against this specification shall be suitable for satisfactory continuous operation under the following tropical conditions.

3.1 Maximum Ambient Air Temperature in shade: 55° C
3.2 Minimum Ambient Air Temperature: (-)10° C
3.3 Maximum Relative Humidity: 95% (non-condensing)
3.4 Minimum Relative Humidity: 10%
3.5 Maximum altitude above mean sea level(MSL) : 3000 m
3.6 Average number of tropical monsoon per annum: 5 months
3.7 Annual Rainfall: 100 mm to 1500 mm
3.8 Maximum Wind Pressure : 150 Kg/Sqm

Smart Meters shall be capable of maintaining required accuracy under hot, tropical and dusty climatic conditions. The Smart Meters shall be suitably designed and treated for normal life and satisfactory operation under hot and hazardous tropical climatic conditions and shall be dust and vermin proof. All the parts and surface, which are subject to corrosion, shall either be made of such material or shall be provided with such protective finish which provides suitable protection to them from any injurious effect of excessive humidity.

4.0 ELECTRICAL SPECIFICATIONS

Description	132 KV	33 KV	11 KV	LT CT operated
Class of Accuracy	0.2S	0.5S	0.5S	0.5S
Type Of Smart Meter	3 Phase 4 Wire			
PT Ratio	(132/√3) kV / (110/√3)V	(33/√3) kV / (110/√3)V	(11/√3) kV / (110/√3)V	240V / 240 V LT
Rated PT secondary voltage (V_{ref})	110 V (P-P), 110/√3 V (P-N) with variation -30% to +20%			415(P-P) : 240 V(P-N) variation -30% to +20%
CT Ratio	200/1A	100/1A	50/5 A	200/5 A
Rated Current (I_b)	1A	1A	5A	5A
Maximum Current (I_{max})	2A	2A	10A	10A
Starting Current	0.1% of I_b at upf			
Rated Frequency	50 Hz ±5%			
Power Factor Range	From Zero lagging through Unity to Zero leading. In Import mode Smart Meter shall be lag only. (Programmable) From Zero lagging to Unity to Zero leading. In Export mode Smart Meter shall be lag only / lag + lead (Programmable)			
Power Loss	Voltage Circuit : <= 1.5W/10VA Current Circuit: <= 1VA			
Resistance to surge voltage of 1.2/50 µSec	8kV peak (minimum)			
Test Voltage at 50 Hz for 1 min	2kV rms			
Clock Time Accuracy	Maximum ±2 min drift per annum	Maximum ±5 min drift per annum		

5.0 TEMPERATURE RISE

IS 14697 as well as 16444 Part II shall have to be followed to ascertain that under normal condition of use, winding and insulation shall not reach a temperature, which might adversely affect the operation of the Smart Meters.

6.0 ACCURACY

Class of accuracy of the meter shall be as per above table (Clause No:4.0)

Limits of error should be in line with IS-14697/16444 Part II.

There shall be no drift in accuracy at least for a period of ten years from the date of supply. In case any drift is noticed which is beyond the permissible limits, the Smart Meter has to be recalibrated / replaced by a new one by the supplier.

7.0 RUNNING AT NO LOAD

Running at no load: When 70% & 120% voltage is applied and no current flows in the current circuit, the test output of the Smart Meter shall not produce more than one pulse.

8.0 CONSTRUCTION

8.1 The case, winding, voltage circuit, sealing arrangements, registers, terminal block, terminal cover & name plate etc. shall be in accordance with the relevant standards. The Smart Meter shall be compact & reliable in design, easy to transport & immune to vibration & shock involved in the transportation & handling. The construction of the Smart Meter shall ensure consistence performance under all conditions especially during heavy rains / very hot weathers. The insulating materials used in the Smart Meter shall be non-hygroscopic, non-ageing & have tested quality. The Smart Meter shall be sealed in such a way that the internal parts of the Smart Meter becomes inaccessible and attempts to open the Smart Meter shall result in viable damage to the Smart Meter cover i.e. break to open type. This is to be achieved by using continuous Ultrasonic welding on all the four sides of the Smart Meter base and cover or any other technology which is either equally or more efficacious.

8.2 The Smart Meter shall comply latest technology such as Microcircuit or Application Specific Integrated Circuit (ASIC) to ensure reliable performance. The mounting of the components on the PCB shall compulsorily be Surface Mounted Technology (SMT) type. Power supply component may be of PTH type. The electronic components used in the Smart Meter shall be of high quality and there shall be no drift in the accuracy of the Smart Meter for at least ten years. The circuitry of the Smart Meter shall be compatible with 16 Bit (or better) ASIC with compatible processor and Smart Meter shall be based on Digital measuring and sampling technique.

8.3 The Smart Meter shall be housed in a safe, high grade, unbreakable, fire resistant, UV stabilized, virgin Polycarbonate casing of projection mounting type. The Smart Meter cover shall be transparent / translucent. But the viewing portion shall be transparent for easy reading of displayed parameters, and observation of operation indicators. The Smart Meter base may not be transparent, but it shall not be black in colour. The Smart Meter casing shall not change in shape, colour, size and dimensions when subjected to 72hrs on UV test as per ASTMD 53. It shall withstand 650 deg.C. glow wire test and heat deflection test as per ISO 75 or as per IEC 60068 -2-5.

8.4 In addition to the above, the Smart Meter cover shall be sealable to the Smart Meter base with at least 2 nos. bar coded seals bearing the identification marks of the Manufacturer. Suitable arrangement shall be made for fitting/fixing of utility seal at two sides of Smart Meter terminal cover in such a manner that any access to the terminal cannot be made possible without removing the seal. There shall also be provision for sealing at the optical port & terminal cover.

8.5 The polycarbonate material of only the following manufacturers shall be used:

8.5.1 **G.E. Plastic :** LEXAN 943A or equivalent, like 943, 123R, 143 for Smart Meter cover & terminal cover / LEXAN 503R or equivalent like 500, 143R, 500R for Smart Meter base and terminal block

8.5.2 **BAYER:** Grade corresponding to above

8.5.3 **DOW Chemical: --do--**

8.5.4 **MITSUBISHI : --do--**

8.5.5 **TEJIN: --do--**

8.5.6 **DUPONT: --do—**

8.6 SMART METER CASE AND COVER

8.6.1 In case, ultrasonic welding using plate / strip is used, the material of plate / strip shall be same as that of cover and base and the strip. The manufacturer's logo shall be embossed on the strip plate. The material of the Smart Meter body (case and cover) shall be of Engineering Plastic.

8.6.2 The Smart Meter cover shall be fixed to the Smart Meter base (case) with Unidirectional Screws, so that the same cannot be opened by use of screwdrivers. These unidirectional screws shall be covered with transparent caps (not required for screw less design), ultrasonically welded with the Smart Meter body and the screw covers shall be embedded in the Smart Meter body in a groove. The Smart Meter shall withstand external magnetic influence as per latest amendments of CBIP Technical Report No.325 including 0.2T AC Magnet, 0.5T Permanent magnet.

8.7 TERMINAL BLOCK AND COVER

8.7.1 The terminal Block, the terminal cover (if not of metal) and the case (if not of metal) shall be of a material which complies with the requirements of IS 11731 (Part-I) method FH 1. The material of which the terminal block is made shall be capable of passing the test given in ISO 75-1 (1993) and ISO 75-2 (1993) for a temperature of 135 °C and a pressure of 1.8 MPa (Method A). The terminal block shall accommodate a conductor having a cross-section at least equivalent to the main current conductors but which a lower limit of $6mm^2$ and shall be clearly identified by earthing symbol.

8.7.2 The terminals may be grouped in a terminal block having adequate insulating properties and mechanical strength. The terminal block shall be made from best quality non- hygroscopic, flame retardant material (capable of passing the flammability tests) with nickel plated brass inserts / alloy inserts for connecting terminals. It shall be rigidly fixed to the base of the Smart Meter so that it cannot be separated from the Smart Meter base without breaking either the Smart Meter base or the terminal block and this fixing arrangement shall be in parallel to the Smart Meter base in such a way that it cannot be viewed or approached from any part of the Smart Meter without breaking the Smart Meter.

8.7.3 The terminals in the terminal block shall be of adequate length in order to have proper grip of conductor. The screws shall have thread size not less than M4 and head having 6 mm. Diameters. The screws shall not have pointed ends at the end of threads. All terminals and connecting screws and washers shall be of tinned / nickel plated brass material. The terminal shall withstand glow wire test at 960 + 15 °C and the terminal shall withstand at least 135 °C, as per IS.

8.7.4 The internal diameter of terminal hole shall be minimum 5.5 mm and center to center distance is 13 mm. The holes in the insulating material shall be of sufficient size to accommodate the insulation of conductor also.

8.7.5 The terminal cover shall be transparent re-enforced Polycarbonate, Engineering Plastic with minimum thickness 2.0 mm and the terminal cover shall be of extended type completely covering the terminal block and, fixing holes. The space inside the terminal cover shall be sufficient to accommodate adequate length of external cables.

9.0 MARKING OF THE SMART METER

Every Smart Meter shall have a nameplate clearly visible, indelible and distinctly marked in accordance with relevant Standards. The following information shall appear on an external plate attached to the Smart Meter cover:

- 9.1 Manufacturer's name or trade mark & place of manufacture
- 9.2 Designation of type
- 9.3 Number of phases & wires
- 9.4 Serial number of Smart Meter
- 9.5 Device ID (as per Table 4 of IS 15959 part 3)
- 9.6 Month & year of manufacture
- 9.7 Principal unit(s) of measurement
- 9.8 PT Ratio
- 9.9 Basic current and rated maximum current
- 9.10 CT Ratio
- 9.11 Reference frequency in Hz
- 9.12 Smart Meter Constant (Impulse/MWh or KWh)
- 9.13 Class index of Smart Meter
- 9.14 Reference temperature
- 9.15 -Property of WBSEDCL
- 9.16 Purchase Order No. & Date
- 9.17 Guarantee (Guaranteed for a period of 5 ½ years from the date of supply)
- 9.18 BIS marking
- 9.19 The sign of Double Square for insulating encased Smart Meters
- 9.20 Barcode/QR Code for Smart Meter serial no. in alpha numeric form, date of manufacture, current rating of the Smart Meter and PO reference, readable by single layer barcode reader
- 9.21 Firmware version

10.0 CONNECTION DIAGRAM AND TERMINAL MARKING

Every Smart Meter shall be indelibly marked with a diagram of connection. For this poly phase Smart Meters, this diagram shall also show the phase sequence for which the Smart Meter is intended. It is permissible to indicate the connection diagram by an identification figure in accordance with relevant standards. The marking of Smart Meter terminals shall appear on this diagram.

11.0 DISPLAY OF MEASURED VALUES

11.1 The Smart Meter shall have alphanumeric display with at least 7 full digit with LCD backlit display, having minimum character height of 10 mm. The data shall be stored in nonvolatile memory. The non-volatile memory ~~shall retain data for a period of not less than 10 years under unpowered condition. Battery back-up memory will~~ not be considered as NVM.

11.2 It shall be possible to easily identify the single or multiple displayed parameters through symbols /legend on the Smart Meter display itself or through display annunciation which shall be self explanatory and symmetric.

11.3 In addition to provide Serial Number of the Smart Meter on the display plate, the Smart Meter serial no. shall also be programmed into Smart Meter memory for identification through communication port for laptop / Smart Meter reading printout.

11.4 Visibility of display in poor light conditions is an important criterion. STN or TN or any better type of advanced

LCD to be used. Proper legends for the displayed parameters to be provided (Factory programmable). Back lit provided for clear visibility shall be uniform throughout all part of the LCD.

11.5 The Smart Meters shall have auto-display mode for pre-selected parameters. Push-Button mode of display shall display all parameters and it shall have priority over auto mode. The Smart Meter shall give clear message on display to indicate that the Smart Meter has experienced tampers and the nature of tamper with date and time of first occurrence, last occurrence and last restoration.

11.6 Connection check, Phase sequence and self diagnostic shall give clear message on display. The Smart Meter shall have a test output (blinking **LED**) accessible from the front and be capable of being monitored with suitable testing equipment. The operation indicator must be visible from the front. Test output device shall be provided in the form of Two separate LED for active and reactive energy with the provision of selecting the parameter being tested (separate LED may also be used with proper distance).

12.0 DISPLAY SEQUENCE (For 11 KV, 33 KV & 132 KV meters)

The Smart Meter shall display the required parameters in two different modes as follows. Display sequence for both auto and Push button must be maintained. no interchange in sequence or display parameter will be accepted. All the display shall have proper legend to identify the same. All the display parameters like auto and push will come under push method. The display parameters as followed are provisional and subject to change after sample testing/firm order. The provision for necessary alteration of display parameters may be made available by the respective manufacturer.

12.1 **Auto Display Mode**: In this mode, the below listed parameters shall be displayed in the following sequence:

12.1.1 LCD Test
12.1.2 Smart Meter Serial Number
12.1.3 Real Time & Date (DD/MM/YYYY)
12.1.4 Rising Apparent Demand with elapsed time while Active Import
12.1.5 Rising Apparent Demand with elapsed time while Active Export
12.1.6 No. of Power Failures
12.1.7 Cumulative Active Import Energy (Cumulative sign/legend must be given)
12.1.8 Cumulative Active Export Energy
12.1.9 Cumulative Reactive Energy – Quadrant-I
12.1.10 Cumulative Reactive Energy – Quadrant-II
12.1.11 Cumulative Reactive Energy – Quadrant-III
12.1.12 Cumulative Reactive Energy – Quadrant-IV
12.1.13 Cumulative Apparent Forward Energy (Active Import)
12.1.14 Cumulative Apparent Forward Energy (Active Export)
~~12.1.15 Apparent Forward Maximum Demand (Active Import)~~
12.1.16 Apparent Forward Maximum Demand (Active Export)
12.1.17 Inst. Voltages – Phase-wise (P-N)
12.1.18 Inst. Currents – Phase-wise
12.1.19 Signed Inst. Power Factor – Phase-wise
12.1.20 Inst. Net Power Factor
12.1.21 Inst. Apparent Power
12.1.22 Signed Active Power in MW
12.1.23 Signed Reactive Power in MVAR
12.1.24 Frequency
12.1.25 Cumulative Billing Count

12.1.26 Cumulative Tamper Count
12.1.27 Cumulative Power Off Hours since manufacturing
12.1.28 Power **OFF** Hours of present month
12.1.29 Phase Sequence & Phase Correspondences of Voltage & Current
12.1.30 Connection Check
12.1.31 Self Diagnosis
12.1.32 Cumulative Net Active Forward Import Energy
12.1.33 Cumulative Net Active Forward Export Energy

12.2 Push Button mode: Over & above the parameters of Auto Display Mode, the following parameters shall be displayed on pressing the push button. The Smart Meter display shall return to auto display mode (mentioned above) if the _push button' is not operated approx. more than 6 seconds.

Import Mode

12.2.1 LCD Test
12.2.2 Smart Meter Serial Number
12.2.3 Real Time & Date (DD/MM/YYYY)
12.2.4 Rising Apparent Demand with elapsed time while Active Import
12.2.5 No. of Power Failures
12.2.6 Cumulative Active Import Energy (Cumulative sign/legend must be given)
12.2.7 Cumulative Reactive Energy – Quadrant-I
12.2.8 Cumulative Reactive Energy – Quadrant-II
12.2.9 Cumulative Reactive Energy – Quadrant-III
12.2.10 Cumulative Reactive Energy – Quadrant-IV
12.2.11 Cumulative Apparent Forward Energy (Import)
12.2.12 Apparent Forward Maximum Demand (Import) With Date & Time
12.2.13 Active Maximum Demand (Import) with date & time
12.2.14 TOD wise Cumulative Active Import Energy
12.2.15 TOD wise Cumulative Apparent Import Energy
12.2.16 TOD wise Apparent Forward Maximum Demand (Import)
12.2.17 TOD wise Active Forward Maximum Demand (Import)
12.2.18 Cumulative Apparent MD (Import)
12.2.19 Cumulative Billing Count
12.2.20 Inst. Secondary Voltages – Phase-wise
12.2.21 Inst. Secondary Currents – Phase-wise
12.2.22 Signed Inst. Power Factor – Phase-wise
12.2.23 Inst. Net Power Factor
12.2.24 Inst. Apparent Power in MVA
12.2.25 Frequency
12.2.26 Signed Active Power in MW
12.2.27 Signed Reactive Power in MVAR
12.2.28 Cumulative Tamper Count
12.2.29 Cumulative Power Off Hours since manufacturing
12.2.30 Power **OFF** Hours of present month
12.2.31 Phase Sequence & Phase Correspondences of Voltage & Current
12.2.32 Connection Check
12.2.33 Self Diagnosis
12.2.34 History 1 Cumulative Active Forward Import Energy
12.2.35 History 1 Cumulative Apparent Forward Import Energy
12.2.36 History 1 Apparent MD while Active Import with Date & Time
12.2.37 Last Billing Date & Time

01/12/23

01/12/23

01/12/23

01/12/23

01/12/23

12.2.38 History 1 Average Power Factor .
12.2.39 Battery Status
12.2.40 Present Tamper Status(PT/CT/Other)
12.2.41 First Tamper Occurrence with Date & Time
12.2.42 Last Tamper Occurrence with Date & Time
12.2.43 Last Tamper Restoration with Date & Time
12.2.44 Cover Open Information with Date & Time
12.2.45 High Resolution Cumulative Forward Active Energy (Import) (2+5 in MWh)
12.2.46 High Resolution Cumulative Forward Reactive Energy – Quadrant-(Q1-Q3)(2+5 in MVArh)
12.2.47 High Resolution Cumulative Forward Reactive Energy – Quadrant-(Q2-Q4)(2+5 in MVArh)
12.2.48 High Resolution Cumulative Forward Apparent Energy (Import) (2+5 in MVAh)
12.2.49 Present Voltage THD
12.2.50 Present Current TDD
12.2.51 Maximum Voltage THD with date & time
12.2.52 Maximum Current TDD with date & time

12.3 Net (Import-Export)

12.3.1 LCD Test
12.3.2 Smart Meter Serial Number
12.3.3 Real Time & Date (DD/MM/YYYY)
12.3.4 Rising Apparent Demand with elapsed time while Active Import
12.3.5 Rising Apparent Demand with elapsed time while Active Export
12.3.6 No. of Power Failures
12.3.7 Cumulative Active Import Energy
12.3.8 Cumulative Active Export Energy
12.3.9 Cumulative Reactive Energy – Quadrant-I
12.3.10 Cumulative Reactive Energy – Quadrant-II
12.3.11 Cumulative Reactive Energy – Quadrant-III
12.3.12 Cumulative Reactive Energy – Quadrant-IV
12.3.13 Cumulative Apparent Forward Energy (Active Import)
12.3.14 Cumulative Apparent Forward Energy (Active Export)
12.3.15 Apparent Maximum Demand (Import) with date & time
12.3.16 Apparent Maximum Demand (Export) with date & time
12.3.17 Active Maximum Demand (Import) with date & time
12.3.18 Active Maximum Demand (Export) with date & time
12.3.19 TOD wise Cumulative Active Import Energy
12.3.20 TOD wise Cumulative Apparent Import Energy
12.3.21 TOD wise Apparent Forward Maximum Demand (Import)
12.3.22 TOD wise Cumulative Active Export Energy
12.3.23 TOD wise Cumulative Apparent Export Energy
12.3.24 TOD wise Apparent Forward Maximum Demand (Export)
12.3.25 Cumulative Apparent MD (Import)
12.3.26 Cumulative Apparent MD (Export)
12.3.27 Cumulative Billing Count
12.3.28 Inst. Secondary Voltages – Phase-wise (P-N)
12.3.29 Inst. Secondary Currents – Phase-wise
12.3.30 Signed Inst. Power Factor – Phase-wise
12.3.31 Inst. Net Power Factor
12.3.32 Inst. Apparent Power
12.3.33 Signed Active Power in MW

12.3.34 Signed Reactive Power in MVAR.
12.3.35 Frequency
12.3.36 Cumulative Tamper Count
12.3.37 Cumulative Power Off Hours since manufacturing
12.3.38 Power **OFF** Hours of present month
12.3.39 Phase Sequence & Phase Correspondences of Voltage & Current
12.3.40 Connection Check
12.3.41 Self Diagnosis
12.3.42 Cumulative Net Active Forward Import Energy
12.3.43 Cumulative Net Active Forward Export Energy
12.2.34 History 1 Cumulative Active Forward Import Energy
12.2.35 History 1 Cumulative Active Forward Export Energy
12.2.36 History 1 Cumulative Apparent Forward Import Energy
12.2.37 History 1 Cumulative Apparent Forward Export Energy
12.2.38 History 1 Apparent MD while Active Import with Date & Time
12.2.39 History 1 Apparent MD while Active Export with Date & Time
12.2.40 Last Billing Date & Time
12.2.41 History 1 Average Power Factor
12.2.42 Battery Status
12.2.43 Present Tamper Status(PT/CT/Other)
12.2.44 First Tamper Occurrence with Date & Time
12.2.45 Last Tamper Occurrence with Date & Time
12.2.46 Last Tamper Restoration with Date & Time
12.2.47 Cover Open Information with Date & Time
12.2.48 High Resolution Cumulative Forward Active Energy (Import) (2+5 in MWh)
12.2.49 High Resolution Cumulative Forward Active Energy (Export) (2+5 in MWh)
12.2.50 High Resolution Cumulative Forward Reactive Energy – Quadrant-I (2+5 in MVArh)
12.2.51 High Resolution Cumulative Forward Reactive Energy – Quadrant-II (2+5 in MVArh)
12.2.52 High Resolution Cumulative Forward Reactive Energy – Quadrant-III (2+5 in MVArh)
12.2.53 High Resolution Cumulative Forward Reactive Energy – Quadrant-IV (2+5 in MVArh)
12.2.54 High Resolution Cumulative Forward Apparent Energy (Import) (2+5 in MVAh)
12.2.55 High Resolution Cumulative Forward Apparent Energy (Export) (2+5 in MVAh)
12.2.56 Present Voltage THD
12.2.57 Present Current TDD
12.2.58 Maximum Voltage THD with date & time
12.2.59 Maximum Current TDD with date & time

12.3. TOD Wise

12.3.1 TOD wise Cumulative Active Forward Energy (Import)
12.3.2 TOD wise Cumulative Active Forward Energy (Export)
12.3.3 TOD wise Cumulative Apparent Forward Energy (Import)
~~12.3.4 TOD wise Cumulative Apparent Forward Energy (Export)~~
12.3.5 TOD wise Apparent MD (Import) with Date & Time
12.3.6 TOD wise Apparent MD (Export) with Date & Time
12.3.7 TOD wise History 1 Cumulative Active Forward Energy (Import)
12.3.8 TOD wise History 1 Cumulative Active Forward Energy (Export)
12.3.9 TOD wise History 1 Cumulative Apparent Forward Energy (Import)
12.3.10 TOD wise History 1 Cumulative Apparent Forward Energy (Export)
12.3.11 TOD wise History 1 Apparent MD (Import) with Date & Time
12.3.12 TOD wise History 1 Apparent MD (Export) with Date & Time

13.0 POWER OFF MODE DISPLAY

The following parameters shall be displayed in Power OFF mode:

13.1 Smart Meter Serial No.
13.2 Real Date &Time
13.3 History 1 Cumulative Active Forward Energy (Import)
13.4 History 1 Cumulative Active Forward Energy (Export)
13.5 History 1 Cumulative Apparent Forward Energy (Import)
13.6 History 1 Cumulative Apparent Forward Energy (Export)
13.7 History 1 Apparent MD while Active Import
13.8 History 1 Apparent MD while Active Export
13.9 Cumulative Billing Count
13.10 Cumulative Tamper Count
13.11 Cumulative Active Import Energy
13.12 Cumulative Active Export Energy

14.0 DISPLAY: OTHER REQUIREMENTS

14.1 List of parameters for Auto, Push Button & Power OFF mode must have the following headers:

14.1.1 Auto Display Mode
14.1.2 Push Button Mode
14.1.3 Power OFF Mode

14.2 Each parameter shall be on Smart Meter display for 10 second the time gap between two auto display cycles shall be 15 sec. instead of 30 sec.

14.3 The register shall be able to record and display starting from zero, for a minimum of 1500 hours, the energy corresponding to rated maximum current at reference voltage and unity power factor. The register shall not roll over in between this duration.

14.4 Push button mechanism shall be of high quality and shall provide trouble free service for a long span of time.
14.5 Up and Down scrolling facility shall be there for Push Button Mode.

15.0 ANTI TAMPER FEATURES

The Smart Meter shall have the following anti-tamper features:

15.1 The Smart Meter shall be capable of recording energy correctly at Import as well as Export mode. If any phase current flows in forward direction, the Smart Meter shall register energy in phase-wise Import counter and if phase current flows in reverse direction, it shall register energy in Export counter. Simultaneous Import/Net (Import – Export) in different phases shall be allowed and this will not be treated as any Tamper event. **Net Register as per pre dominance to be provided.**

15.2 The Smart Meter shall work correctly irrespective of phase sequence of supply. (There must be an indication in display & down loaded data).

15.3 The Smart Meter shall work correctly even in absence of neutral. For reference voltage Vref between 70% to 120 %, accuracy must be maintained within permissible limits as per IS.

15.4 The registration shall not be affected more than + 4% if high frequency (55Hz to 100Hz) or low frequency (45Hz to 30 Hz) AC signal w.r.t. earth is applied to the Smart Meter neutral. Smart Meters which are immune or will maintain better accuracy, will be preferred.

15.5 The Smart Meter shall be immune to Electro Static Discharge or Sparks of 35 KV (approx) induced by using frequency-generating devices having very high output voltage.

Tests in this respect will be conducted by using commonly available devices and during spark discharge test, spark will be applied directly at all vulnerable points of the Smart Meter for a period of 10 minutes (at an interval of 1minute (approx) between two consecutive strokes) and Smart Meter shall maintain accuracy after the test under this condition. Accuracy will be checked during and after application of spark discharge Test. Smart Meter shall record correctly within the specified limits of errors. Beyond 35 KV the Smart Meter shall record tamper if not immune.

THD

- For any consumer crossing the threshold of THD occurrence and restoration with date and time is to be registered as Tamper Events. Manufacturer may have their own logic to implement this THD threshold limits breach by the Consumer.

- This THD threshold limits breach data should be availabe as Event data with time stamping The meter shall record the total energy i.e the sum of Fundamental Energy at 50 Hz and Harmonic Energy for active as well as reactive parts.
- Provision for displaying the THD and TDD (Total Demand Distortion) as a percentage (%) of voltage and load current (IL) respectively in each phase.
- THD and TDD limits due to harmonics should be strictly as per IEEE Standard 519-1992 and should be < 0.05 in each case., assuming (ISC/IL) <= 20 for TDD as per IEEE-519-1992.
- The breach in THD and TDD limits should be logged as an event and the meter should generate a flag whenever the threshold (user configurable) of 5% is exceeded..
- THD & TDD values should have 15 minutes Integration period in Load Survey.
- Accuracy of harmonics recording shall be as per meter accuracy class.
- The meter shall record and display total forwarded energy including forwarded Harmonic energy in following conditions.

(This is special requirement of the WBSEDCL)
(a) Voltage and current both in phase
(b) Voltage and current both out phase
(c) Voltage in phase and current out phase
(d) Voltage out phase and current in phase

Above tests shall be carried out at
Ref. Voltage, 0.5 Imax and UPF with
10% 5th harmonics in voltage and 40% 5th harmonics in current.

In all above conditions, meter shall record total energy = 1.04 (± 0.005) times fundamental energy.

% Maximum THD & TDD with date and time since Last Reset should be made available Threshold Values of all above occurrence and restoration are attached herewith. Snapshot values of Phase Voltage. Phase Current & Phase wise Power Factor, Active Energy value during occurrence & restoration to be provided in all the above mentioned tamper conditions in BCS AND HES with date and time. (Event logging Snapshots shall be considered when the actual phenomenon occurred). The logging time for recording occurrence and restoration of all tamper events except Magnetic & Neutral Disturbance tamper, shall be 5 min. Magnetic tamper shall appear instantaneously, Neutral Disturbance within 3 min.

****

15.6 The Smart Meter shall be capable of recording occurrence and restoration with date and time in respect to the following tamper events:

15.6.1 Power failure (Tamper count not to be increased) - as per tamper logic
15.6.2 Invalid Voltage- as per tamper logic
15.6.3 Missing Potential (phase wise) —as per tamper logic
15.6.4 High Voltage – as per tamper logic
15.6.5 Voltage Unbalance – as per tamper logic
15.6.6 CT Open- as per tamper logic
15.6.7 CT Bypass/ CT Short - as per tamper logic
15.6.8 CT Unbalance - as per tamper logic
15.6.9 Over Current - as per tamper logic
15.6.10 Neutral Disturbances (If it is logged) - as per tamper logic
15.6.11 Magnetic Disturbances - as per tamper logic

15.7 Threshold Values of all above occurrence and restoration are provided with in annexed tamper logic sheet.

15.8 The logging time for recording occurrence and restoration of all tamper events except magnetic & neutral disturbance tamper, shall be 5 minutes. Magnetic tamper shall appear instantaneously and neutral disturbance within 3 minutes. However, it shall be programmable.

15.9 Snapshot values of Phase Voltage, Phase Current & Phase wise Power Factor, Active Energy value during occurrence & restoration shall have to be provided in all the above mentioned tamper conditions in BCS AND HES with date and time.(In Event logging Snapshots shall be considered to be reflection of the actual phenomenon occurred).

15.10 All authenticated commands shall be Base Computer Software (BCS AND HES) controlled. All transactions with Smart Meter shall be date and time logged, in the downloaded data (Last 12 month's transactions).

15.11 Properly designed Smart Meter tamper logic shall be provided and clearly explained in the bid. The tamper logic shall be capable of discriminating the system abnormalities from source side and load side and it shall not log/record tamper due to any source side abnormalities. More than one tamper *CT related/ PT related/ others* shall not be logged at a time. A minimum of 300 events (one event means either occurrence or restoration) of all types of tamper with date & time stamping shall be available in Smart Meter memory compartment wise. The logging will be on FIFO basis. The events will be divided into three compartments like CT related (148 Events), PT related (88 Events) and others (64 Events).

15.12 Smart Meter shall have a continuous and clear indication in its display if top cover is removed / opened and even re-fixed (non rollover) and only cover open must be logged in BCS AND HES without any restoration. COVER OPEN tamper is to be displayed after every para Smart Meter displayed in Auto Display Mode.

16.0 INFLUENCE QUANTITIES

The Smart Meter shall work satisfactory with guaranteed accuracy as per limit provided in IS: 14697/16444 Part II under presence of the following quantities:

16.1 Electromagnetic field
16.2 External magnetic field
16.3 Radio frequency interference
16.4 Vibration
16.5 Harmonic wave form
16.6 Voltage fluctuation
16.7 Frequency variation
16.8 Electromagnetic high frequency field
16.9 Electrostatic discharge

17.0 IMMUNITY TO ELECTRO MAGNETIC DISTURBANCE

The Smart Meter shall be designed in such a way so that external electromagnetic field upto 0.5Tesla or electrostatic discharge up to 35kV either shall not influence its performance .i.e. the Smart Meter shall remain immune to it, or, if not so, it shall log event and shall record energy with I_{max}.

18.0 MEASUREMENT OF HARMONICS

The Smart Meter shall be capable of measuring fundamental as well total energy i.e., fundamental plus harmonics energy. Total energy (both active & apparent) shall be made available on Smart Meter display. Provision for utilizing measured fundamental energy shall be kept for future use. Both total and fundamental energy shall be logged in the Smart Meter memory and shall have provision for downloading to the BCS AND HES through the laptop.

19.0 MAXIMUM DEMAND & ITS RESETTING

19.1 The Smart Meter shall be capable of recording the Apparent MD with integration period of 15 minutes (programmable).

19.2 The Smart Meter shall also record MD at preset dated and time.

19.3 MD reset shall be through each of the three means:

- 19.3.1 Automatic resetting at preset date & time (at present it will be at 00.00 hrs of the first day of every month)
- 19.3.2 Manually i.e., by using push button
- 19.3.3 Through Remote Communication command.

19.4 The means by which the reset has been done shall be made available in downloaded data.

19.5 MD reset button shall have proper sealing arrangement.

19.6 There shall be separate Push button for scrolling display (up and down) and MD reset. If only two Push buttons are used minimum 30 sec pressing is required for MD reset.

20.0 LOAD SURVEY

The Smart Meter shall be capable of recording load survey (both in Import/Net (Import - Export) mode) of Active Energy, Reactive Energy, Apparent Energy, Active Demand, Apparent Demand, Phase wise Current , Phase wise Voltage & Power Factor for a period of minimum 45 days for 15 minute integration period in both tabular as well as graphical format.

20.1 Active & Apparent Energy in MWh/KWh & MVAh/KVAh as the case may be(Import & Export mode shall be shown clearly)

20.2 Reactive Energy in MVARH/KVarh as the case may be (Import & Export mode shall be shown clearly)

20.3 Demand in MVA/KVA and MW/KW as the case may be (Import & Export mode shall be shown clearly)

20.4 Current – phase-wise

20.5 Voltage – phase-wise

20.6 Power Factor

It shall be possible to transfer this data to BCS AND HES directly from Smart Meter or through MODEM or through GSM/CDM/GPRS communication. The data so obtained shall be displayed in both graphical & tabular format in BCS AND HES.

21.0 TIME OF DAY FACILITIES

The Smart Meter shall have facilities to record Active, Apparent Energies and MD in at least 8 zones. The time zones shall be user programmable through authenticated Laptop/RMR command. Necessary software for the same is to be provided by the Smart Meter supplier .At present TOD timings will be programmable as follows:

21.1 TOD 1: 06:00 Hrs to 17:00 Hrs.

21.2 TOD 2: 17:00 Hrs to 23:00 Hrs.

21.3 TOD 3: 23:00 Hrs to 06:00 Hrs.

22.0 SMART METER READING DURING POWER OFF

In absence of input voltages, with the help of internal battery or external battery pack Smart Meter data shall be displayed in power off mode. It shall also be possible to read & download data using Laptop. In case of external battery pack battery the arrangements shall be such that hands free operation is possible. In case of external battery 10 years guarantee must be given for the external battery pack. Separate battery shall be used for this purpose (not RTC or processor battery). In case of Lithium battery rating shall be more than 500mAh.

23.0 SELF DIAGNOSTIC FEATURES

The Smart Meter shall be capable of performing complete self-diagnostic check to monitor the circuits for any malfunctioning to ensure integrity of data in memory location. The details of malfunctioning shall have to be recorded in the Smart Meter memory. The bidder shall furnish the details of self-diagnostic capability features, viz Memory (NVM) status, Battery status, RTC Status etc. and it shall have to be made available in display.

24.0 COMMUNICATION CAPABILITY:

24.1 The Smart Meters will have the capability on smart features with GPRS Connectivity under the scope of IS 16444 Part II/15959.

24.2 The Smart Meter shall have a galvanic isolated Optical Port as per IEC 1107/ANSI/PACT so that it can be easily connected to a handheld Common Smart Meter Reading Instrument Laptop/PC for data transfer.

24.3 The optical port shall be provided with proper sealing arrangement so that its cover can't be opened without breaking its seal. In case sealing arrangement is not there, access through authentication shall have to be ensured.

24.4 A Serial Port (RS485 or **RJ11**) shall have to be provided inside the terminal cover to enable automatic Smart Meter reading through modem, if required in future. The Serial Port shall be housed inside the Smart Meter terminal cover so that it can't accessed without opening the sealed terminal cover.

24.5 The stored data in the Smart Meter shall be available through Laptop even when the display of the Smart Meter is not available.

24.6 Date in the Smart Meter shall be reset only through commands from the Laptop. Correction of RTC time, change of TOD timings etc. shall be done through Laptop utilizing authenticated command set by BCS AND HES.

24.7 Billing parameters shall be factory programmable.

24.8 Meter shall have ability to communicate with HES on any one of the technologies mentioned in IS 16444 (RF/Cellular/PLC) in a secure manner. For this project only, cellular based communication shall be used and the meter shall accommodate dual SIM Card or e-SIM of any service provider to meet the Service Level Agreements (SLAs). The meter shall log the removal of the plug-in type communication module removal /nonresponsive event with snapshot.

24.9 Plug-in Communication Module(NIC): The smart meter shall have a dedicated sealable slot for accommodating plug-in type bi-directional communication module which shall integrate the Cellular communication technology with the smart meter and act as interface between the meter and HES. The Plug-In module shall be field swappable/replaceable.

24.10 Integration: The bidder must ensure bi-directional data communication between the meter and HES for cellular communication technology.

24.11 Software and support: 1. The bidder shall supply following software and provide required training & manuals to use the same free of cost:

Software for local communication, i.e., for Mobile App, Laptop and PC. This software shall be compatible with Android (11.0 and above) or Windows OS (10 and above), whichever is applicable.

Software for firmware upgrading from remote-end mass deployment.

2. Bidder shall ensure integration of software with any of WBSEDCL system during the life of meter free of cost. WBSEDCL will provide all the

required support for integration activity.

3. The bidder shall have to ensure DLMS compliance of the meter for both local &remote communication through its Optical port and NIC card.

24.12 Software for local communication: The manufacturer has to provide software capable of downloading all the data stored in meter memory through Mobile app, Laptop and PC. The manufacturer shall also provide software for Android and Windows devices. The bidder has to ensure that all required reading and diagnostic tools are available with them to ensure SLA as mentioned in Tender. Specification and costing of these tools shall be shared with WBSEDCL incase WBSEDCL wants to maintain their own devices.

24.13 Meter data for manual collection through mobile app: In case the meter data is not received through AMI, the manual meter reading data using mobile devices through the meter reading app shall be directly uploaded to the HES.

24.14 Training: Manufacturer shall impart training to WBSEDCL personnel for usage of software.

24.15 Port protection: All ports shall be optically isolated from the power circuit.

24.16 Operation: Optical port and NIC Card shall work independently. Failure of one (including display) shall not affect the working of the other.

24.17 Communication protocol : As per IS-15959(Part3) as applicable.

24.18 Data transfer rate: Communication ports shall have to support data transfer rate of 9600 Bps (minimum).

24.19 Data Security: Advanced Security outlined in clause 7.1.2 of IS-15959(Part1)

24.20 Encryption for data communication: As per clause 7.1 of IS15959 (Part2)

24.21 Encryption/ Authentication For data transport : As perclause7.2 ofIS15959 (Part2)

24.22 Key requirements and handling: As perclause7.3 of IS15959 (Part2)

24.23 IP communication profile support: As per clause 8 of IS 15959(Part2).
Note: Meter shall support TCP-UDP/ IP communication profile for smart meter to HES.

24.24 Antenna port: The meter should have a port for connecting external antenna of strength 5 dBi or more in case the signal strength of the SIMs is poor. The antenna shall be supplied by the manufacturer for this scenario.

25.0 General requirement for common pluggable communication module for Smart Meters

Part I

1. Recommended Module Placement location: In order to improve the performances of Cellular technologies encompassing, it is recommended to place the communication module anywhere on the accessible part of the meter. This will also enable an easy approach to improve antennae performances.
2. Meter shall have the means of tamper detection to record the event(s) of the removal of the communication module set from the meter, irrespective of whether the meter is in power on (has supply) or powered off (no supply) condition.
3. The Module shall be hot swappable and shall fit snugly inside the meter box, so that the same IP class of the meter is maintained.
4. A transparent cover may be used for the purpose.

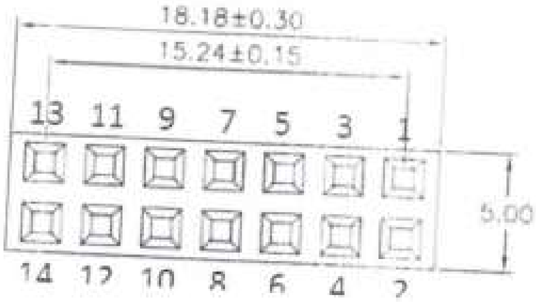
a. To have a sealing arrangement with the meter body as well as,
b. For easy viewing of LED indicators and antenna assembly without having to open the cover.

Part II

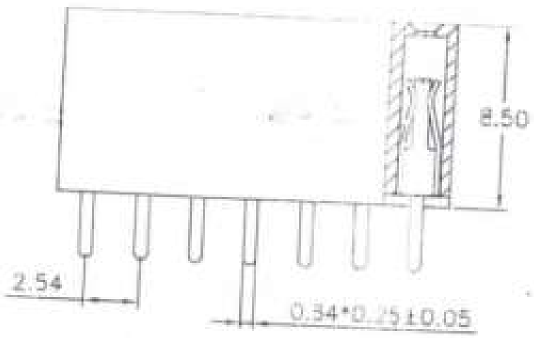
Communication interface: The meter shall have a slot of an appropriate size to allow for the pluggable communication to be fit in to the meter. The meter shall provide a 14-pins Female socket connector (2*7pin, 2.54mm). The socket shall be selected and positioned to ensure that the male pins on the communication module can connect reliably and easily connect with the female contactors on the meter.

Female connector

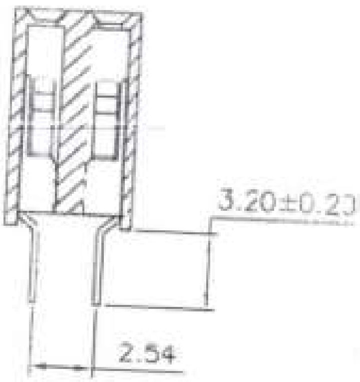
1. Front View :



2. Top View:



3. Side View:



PIN Outs may be provided as per below details:

Pin No	Name	Input/output	Description

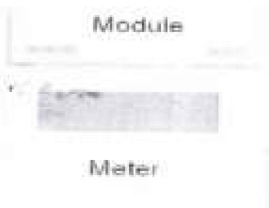
1	Reserved	/	/
2	Reserved	/	/
3	Power EN	Output	Control the module's power supply
4	Reserved	/	/
5	Reserved	/	/
6	Meter TXD	Output	To Module UART port RXD, Min.38400
7	Meter RXD	Input	From Module UART port TXD, Min.38400
8	Reserved	/	/
9	RTS	Input	Input digital signal from module
10	RST	Output	Reset signal for module
11	CTS	Output	Output digital signal to module
12	+Vdc	Power	As per IS16444
13	GND	Common	Ground Reference Potential
14	GND	Common	Ground Reference Potential

Part III

The following reference size may be adhered to irrespective of a single or multiple communication options provisioned on the same module. This standard form factor and dimensions will enable physical and functional interoperability with different makes of meters.

A. Module 3-D views (For Representational Purpose Only)

1. Module in meter (Top View)



2. 3D View

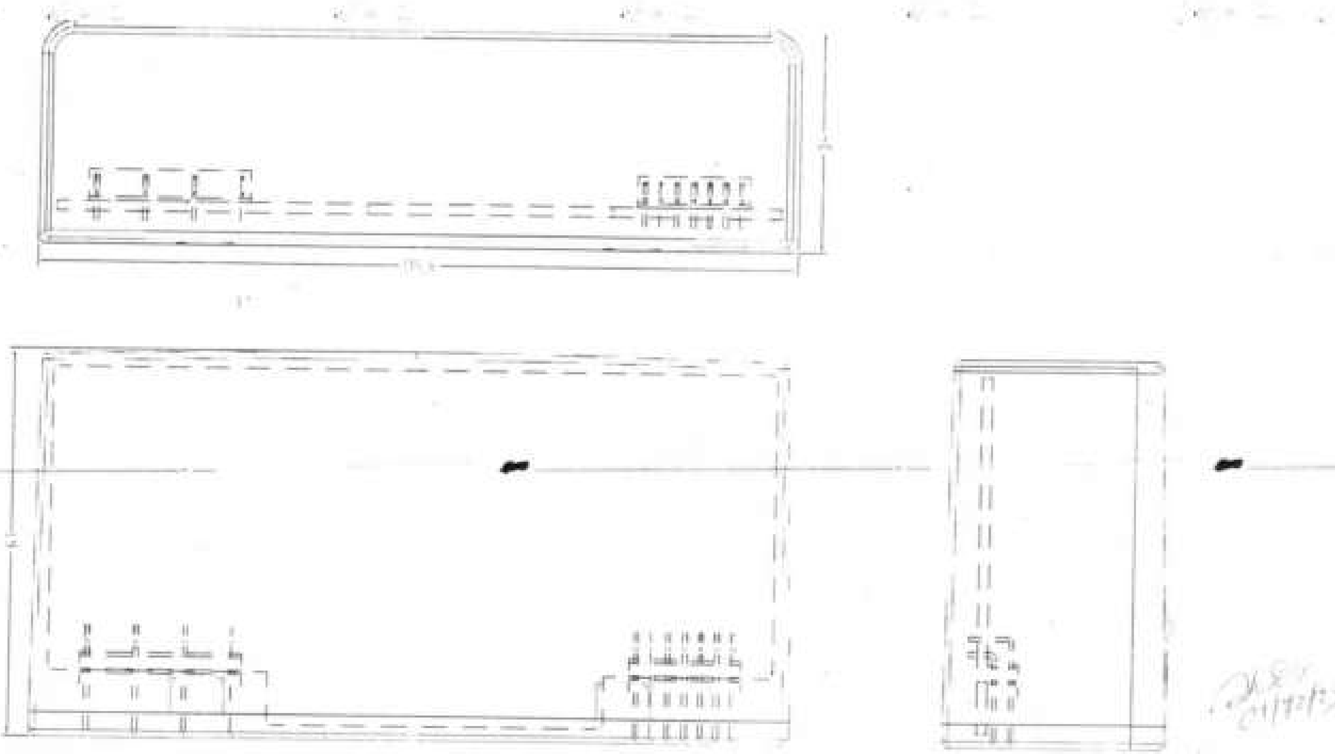
3.Front View4. Back View

5.Side View

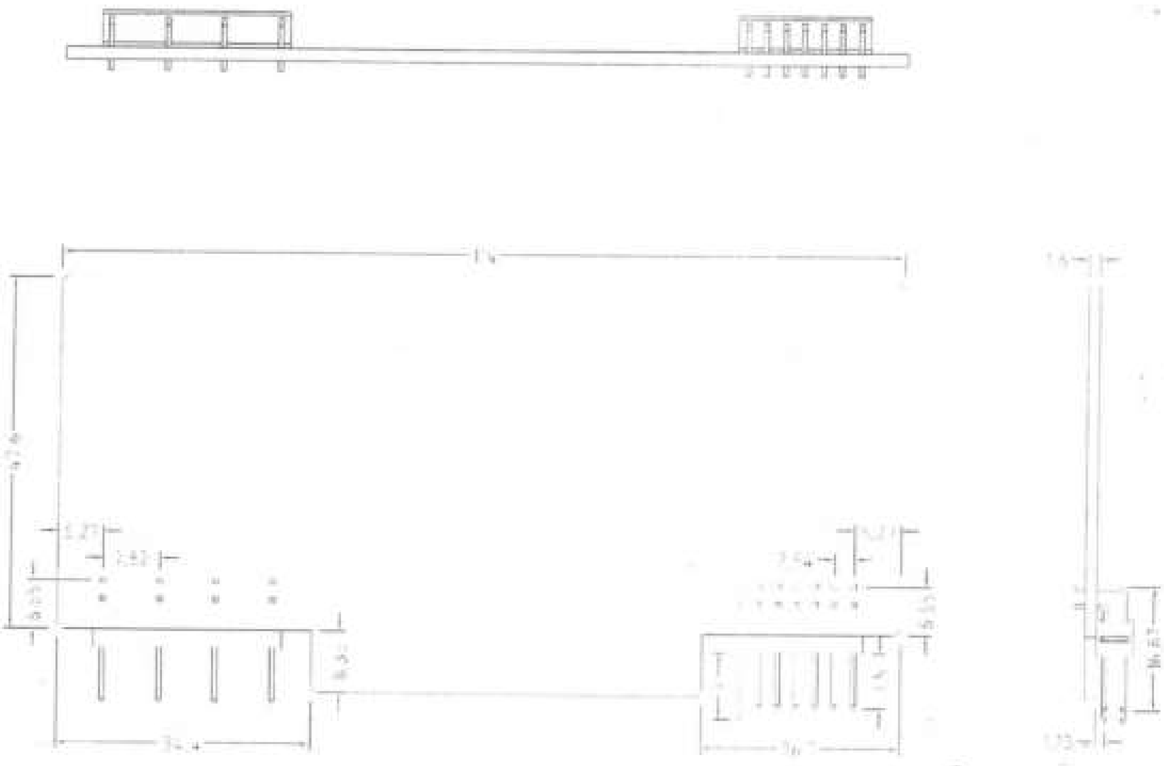
6. Top View

7.Bottom View

B. Module Dimensions



C. Overall view of the module's PCBA:



Notes: Module Reference Sizes: unit

26.0 TECHNICAL SUPPORT, MANUALS &TRAINING

Extensive technical support, detailed technical literature (shall supply with each Smart Meter at the time of packing) & training is to be provided by the manufacturer free of cost.

In case external battery packs are required for the offered Smart Meters, those are to be provided by the manufacturer and shall have clear mention in offered bids.

27.0 BASE COMPUTER SYSTEM & SOFTWARE REQUIREMENTS

27.1 The data stored in the Smart Meter memory including **defrauded energy** shall be available on the BCS AND HES.

27.2 BCS AND HES shall give all details pertaining to billing and load survey data.

27.3 All the data available in the Smart Meter including energy, MD etc. with date and time stamp, new TOD time zones and historical data shall be available in BCS AND HES after down loading. All the data/ items of Push Button Mode Display Parameter List should be available in BCS / HES in preferably better resolution.

27.4 The Smart Meter condition details shall also be transferred into the BCS AND HES including abnormal voltage & current conditions, tamper events etc.

27.5 Facility to view data incorporating external multiplying factor due to installed CT & PT shall have to be provided in BCS AND HES.

27.6The BCS AND HES shall have facility to convert Smart Meter reading data into user definable ASCII file format so that it can be integrated with the billing system or any other third party software. The user shall have the flexibility to select the parameters to be converted into ASCII file.

27.7 The bidder has to supply the Smart Meter reading protocol and API free of cost. The bidder shall indicate the relevant standard to which the protocol is compliant.

27.8 In BCS AND HES twelve months' data for MWh, MVAh, MD in MVA (total & TOD wise), average load factor, average power factor in both Import/Net (Import - Export) mode must be available.

27.9 Six copies of BCS AND HES shall be provided for downloading data and issue of authenticated command from Laptop.

27.10 Smart Meter Data Display:

27.10.1 The BCS AND HES shall show electrical conditions existing at the time of reading the Smart Meter in tabular forms as well as in graphical format (phase or diagram)

27.10.2 All the information shall be shown in a manner which user can understand easily.

27.10.3 All the load survey data shall be available in numerical as well as graphical format. It shall also be possible to view this data in daily, weekly and monthly formats. The load survey graph shall also show values where the cursor/pointer is placed for selected or all parameters.

27.10.4 All the information about tamper events shall be accompanied with date & time stamping along with the -Snapshot(details) of the respective electrical conditions. This information shall be displayed in the sequence in which it happened, in cumulative format as well as summary format. The cumulative format shall segregate a particular tamper information and summary report should show count of tamper occurrence and the duration for which Smart Meter remained under tamper condition.

27.11 Support Display: There shall be user-friendly method for viewing current and stored history Smart Meter data. All information about a particular consumer shall be segregated and available at one place so that locating any consumer's past data is easy. It shall be possible to locate/retrieve data on the basis of one of the following particulars:

i. Consumer ID
ii. Smart Meter Serial No.
iii. Date of Smart Meter Reading
iv. Location

27.12 Configurability: It shall be possible to get selective print out of all the available data of the Smart Meter. Print out shall not include anything and everything available with the BCS AND HES. The software shall support "Print Wizard" or similar utility whereby user can decide what to print out. The user of the software need not revert back to the supplier of the software for modifying the software just to print what he desires.

It is very important that the BCS AND HES has the feature to export available data to ASCII or spreadsheet format for integrating with the WBSEDCL's billing system. Here again an "Export Wizard" or similar utility shall be available whereby user can select file format (for ASCII or for spreadsheet), what data to export, the field width selection (whether 8 characters or 10 characters, to include decimal point or not, number of digits after decimal point) etc.

27.13 Security: The BCS AND HES shall have multi level password for data protection and security. The first level shall allow the user to enter the system. The different software features shall be protected by different passwords. The configuration of passwords shall be user definable. The software installed on one computer shall not be copy-able on to another computer.

27.14 Help: The exhaustive on-line Help should be available with the software so that user can use all the features of the software by just reading the Help contents.

28.0 Daily Load Profile Parameter:

01/12/23 01/12/23 01/12/23 01/12/23 01/12/2023 01/12/23

1. Real Time & Date (DD/MM/YYYY)
2. Cumulative Active Import Energy
3. Cumulative Active Export Energy
4. Cumulative Apparent Forward Energy (Active Import)
5. Cumulative Apparent Forward Energy (Active Export)
6. Apparent Forward Maximum Demand (Active Import)
7. Apparent Forward Maximum Demand (Active Export)
8. Active Forward Maximum Demand (Import)
9. Active Forward Maximum Demand (Export)
10. TOD wise Cumulative Active Forward Energy (Import)
11. TOD wise Cumulative Active Forward Energy (Export)
12. TOD wise Cumulative Apparent Forward Energy (Import)
13. TOD wise Cumulative Apparent Forward Energy (Export)
14. TOD wise Apparent MD (Import) with Date & Time
15. TOD wise Apparent MD (Export) with Date & Time

The above parameters shall be logged at midnight (at 00:00 hrs) and these parameters are meant for billing purpose. The storage time will be same as load survey data.

29.0 GENERAL REQUIREMENTS

29.1 GUARANTEED TECHNICAL PARTICULARS: The bidder shall furnish all the necessary information as desired in the Schedule of Guaranteed Technical Particulars and data, appended with this Specification. If the bidder desires to furnish any other information in addition to the details as asked for, the same may be furnished against the last item of this Annexure– I.

29.2 TECHNICAL DEVIATIONS: Any deviation in Technical Specification as specified in the Specification shall be specifically and clearly indicated in the Schedule of deviation format.

29.3 TESTING:

29.3.1 Type Testing of Smart Meter: The offered Smart Meters shall be type tested at any NABL accredited laboratory in accordance with relevant IS and CBIP Report 325 with their latest amendments. The type test report shall not be more than 3 (three) years old. A copy of the Type Test results shall be enclosed with the offer. If there is any modification in the design Parameters of the specifications or use of constituent materials in the offered Smart Meters submitted with the offer, from the Smart Meter which was submitted type tested, which may affect the characteristics as well as parameters of the Smart Meter, revised type test certificates as per the design, parameters and constituent material used in the offered Smart Meter, shall have to be submitted failing which the offer may be liable to be rejected. Type Test Certificate from any NABL accredited Lab shall only be considered. Type test certificate shall contain the following information clearly:

29.3.1.1 Type of display or LCD
29.3.1.2 Class of accuracy
29.3.1.3 Smart Meter constant
29.3.1.4 Type of Smart Meter

29.3.2 Acceptance tests: The acceptance tests as stipulated in CBIP / IS (with latest amendments) shall be carried out by the supplier in presence of purchaser's representative. In case of failure of Smart Meters as specified in Recommended Sampling Plan of IS-14697/ 16444 Part-II, the entire lot will be treated as rejected. Also the following additional tests are to be carried out on one Smart Meter randomly selected from each lot offered for inspection / acceptance testing. In case of failure of any single Smart Meter the entire lot will be rejected.

29.3.2.1 Magnetic induction of external origin (AC & DC)

29.3.2.2 Tamper & Fraud protection, as per relevant Clause of this specification.

29.3.2.3 Test of endurance upto 150% of I_{max}, for two hours, followed by verification of limits of error.

29.3.2.4 Verification of internal components.

29.3.2.5 Dry Heat Test under Test of Climatic Influences in relevant IS of one Smart Meter from the offered lot is to be arranged by the supplier at any NABL accredited laboratory, at his cost.

29.3.3 Routine Tests : Each and every Smart Meter of the offered lot shall undergo the routine tests as well as functional tests as per IS: 14697/1999, CBIP Report 325, 16444 Part-II and after sealing the Smart Meters, the manufacturers will have to submit the routine test report of all the Smart Meters as well as a statement showing seal Sl. Nos. against each Smart Meter Sl. No. of offered lot in soft copy (MS WORD or EXCEL format), to the Chief Engineer(Procurement and Contract) and the Chief Engineer(DTD), along with offer letter for acceptance test.

29.4 TEST FACILITIES: The tests for equipment / instrument shall be carried out as per relevant Standards and test certificates shall be furnished for scrutiny. The Bidder shall indicate the details of the equipment available with him for carrying out the various tests as per relevant Standards. The bidder shall indicate the sources of all equipments / instruments.

29.5 The standard Smart Meters used for conducting tests shall be calibrated periodically at any NABL Accredited Test Laboratories and test certificates shall be available at Works for verification by purchasers' representative.

29.6 The manufacturer shall have at least the following testing facilities to ensure accurate calibration:-

29.6.1 AC high voltage test

29.6.2 Insulation test

29.6.3 Test of no load condition

29.6.4 Test of Starting condition

29.6.5 Test on Limits of error (Automatic Testing facility with ICT)

29.6.6 Power loss in voltage and current circuit

29.6.7 Test of Repeatability of error

29.6.8 Test of Smart Meter constant

29.6.9 Test of magnetic influence (As per CBIP 325 & Permanent Magnet)

29.7 INSPECTION:

29.7.1 The purchaser may carry out the inspection at any stage of manufacture. The manufacturer shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipment under this specification by the purchaser shall not relieve the supplier of his obligation of furnishing the equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective. All acceptance tests and inspection shall be made at the place of manufacturer unless otherwise especially agreed upon by the Bidder and purchaser at the time of purchase. The Bidder shall provide all reasonable facilities without charge to the inspector, to satisfy him that the equipment is being furnished in accordance with this specification.

29.7.2 The supplier shall keep the purchaser informed in advance, about the manufacturing programme for each lot so that arrangement can be made for inspection. The purchaser reserves the right to insist for witnessing the acceptance / routine testing of the bought out items. The supplier shall give 15 days for local supply / 30 days in case of foreign supply advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests.

29.7.3 The purchaser reserves the right to get type test any Smart Meter, for Smart Meter casing etc. from any of the offered lots, reserve at any destination stores.

30.0 SUBMISSION OF SAMPLE SMART METER

30.1 The bidder will submit his sample Smart Meters in sealed casing / cartoon along with relevant Smart Meter documents (**As per Annexure-IV**), on any working day, from 11.00 A.M. to 04.00 P.M. on weeks days & from 11.00 A.M. to 01.00 P.M. on Saturday within the specified period of submission latest by 01.00 P.M. on the last day of submission of bid to the Office of the Chief Engineer (DTD), Abhikshan, Sec-V, Salt Lake, Kolkata-91. The bidder will be given a receipt, jointly signed by the bidder and DTD officials, mentioning the samples and papers submitted by the bidder as per check list.

30.1.1 While submitting the samples and required documents as per Annexure-IV, the bidder shall submit three numbers of sealed Smart Meters as per the specifications stated herein before.

30.1.2 The date of testing of sample Smart Meters will be intimated to the bidders by CE(DTD) and during such test other bidders will also be allowed to witness the testing. Sample submission and Test procedure may be changed due to emergency requirement. On the date of testing of sample Smart Meters of a particular bidder, he shall come prepared with the following.

30.1.2.1 BCS AND HES (as per specification)
30.1.2.2 Laptop compatible with BCS AND HES and loaded with Laptop software.
30.1.2.3 Modem and accessories for testing the remote Smart Meter reading
30.1.2.4 Any other accessories required for observing the performance and capabilities of the Smart Meters.

31.0 QUALITY ASSURANCE PLAN

The design life of the Smart Meter shall be minimum 10 years and to prove the design life the firm shall have at least the following quality Assurance Plan:

31.1 The factory shall be completely dust proof.
31.2 ~~The test rooms shall be temperature and humidity controlled as per relevant Standards.~~
31.3 All test equipment shall have their valid calibration certificates.
31.4 Smart Meter shall be tested in fully automatic test bench with ICT and results shall be printed directly without any human errors.
31.5 Power supplies used in test equipment shall be distortion free with sinusoidal wave forms, maintaining constant voltage, current and frequency as per the relevant Standards.

32.0 THE CHECKS TO BE CARRIED OUT DURING MANUFACTURING OF THE SMART METERS

32.1 Smart Meter frame dimensions tolerances shall be minimal.
32.2 The CT coil shall be made totally encapsulated and care shall be taken to avoid ingress of dust and moisture inside the coil.

32.3 The assembly of parts shall be done with the help of jigs and fixtures so that human errors are eliminated.

33.0 LAB FACILITY: The laboratory of manufacturer must be well equipped for testing of the Smart Meters. They must have computerized standard power source and standard equipment calibrated not later than a year (or as per standard practice).

34.0 MANUFACTURING ACTIVITIES:

34.1 All the materials, electronics and power components, ICs used in the manufacture of the Smart Meter shall be of highest quality and reputed make to ensure higher reliability, longer life and sustained accuracy. The manufacturer shall use Application Specific Integrated Circuit (ASIC) or Micro controller for Smart Metering functions.

34.2 The electronic components shall be mounted on the printed circuit board using latest Surface Mounted Technology (SMT) except power components by deploying automatic SMT pick and place machine and re flow solder process. The electronic components used in the Smart Meter shall be of high quality and there shall be no drift in the accuracy of the Smart Meter at least up to 10 years.

34.3Further, the Bidder shall own or have assured access (through hire, lease or sub-contract) of the mentioned facilities. The PCB material shall be of glass epoxy FR-4 grade conforming to relevant standards.

34.4 All insulating materials used in the construction of Smart Meters shall be non-hygroscopic, non-ageing and tested quality. All parts that likely to develop corrosion shall be effectively protected against corrosion by providing suitable protective coating. Quality shall be ensured at the following stages.

34.4.1 At PCB manufacturing stage, each board shall be subjected to bare board testing.

34.4.2 At insertion stage, all components shall undergo testing for conforming to design parameters and orientation. Complete assembled and soldered PCB shall undergo functional testing using test equipments (testing jig).

34.5 Prior to final testing and calibration, all Smart Meters shall be subjected to accelerated ageing test to eliminate infant mortality, i.e., Smart Meters are to be kept in ovens for 72 hours at 55 deg Centigrade temperature & atmospheric humid condition. After 72 hours Smart Meters shall work correctly. Facilities / arrangement for conducting ageing test shall be available with the manufacturer.

34.6 The calibration of Smart Meters shall be done in-house.

35.0 DOCUMENTATION:

33.1 Twenty sets of operating manuals shall be supplied to the office of the CE (DTD) for distribution at sites.

33.2 One set of routine test certificates shall accompany each dispatch consignment.

33.3 The acceptance test certificates in case pre-dispatch inspection or a routine test certificate in cases where inspection is waived shall be approved by the purchaser.

36.0 GUARANTEE

34.1 The Smart Meters shall be guaranteed for a period of **5½ years** from the date of supply.

34.2 Life of battery used for the Smart Meter shall be guaranteed for 10 years.

37.0 REPLACEMENT OF DEFECTIVE SMART METERS

The Smart Meters declared defective within the above guarantee period by the WBSEDCL shall be replaced by the supplier up to the full satisfaction of the WBSEDCL at the cost of supplier within one month on receipt of intimation. Failure to do so within the time limit prescribed shall lead to imposition of penalty of twice the cost of Smart Meter. The same may lead to black listing even, as decided by WBSEDCL. In this connection the decision of WBSEDCL shall be final.

38.0 PACKING & FORWARDING

38.1 The equipment shall be packed in cartons / crates suitable for vertical / horizontal transport as the Case may be, and suitable to with stand handling during transport and outdoor storage during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing. The easily damageable material shall be carefully packed and marked with the appropriate caution symbol. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Supplier without any extra cost shall supply any material found short inside the packing cases immediately.

38.2The packing shall be done as per the standard practice as mentioned in IS 15707: 2006. Each package shall clearly indicate the marking details (for e.g. manufacturer's name. Sl. Nos. of Smart Meters in the package, quantity of Smart Meter, and other details as per supply order). However, the supplier shall ensure the packing is such that, the material shall not get damaged during transit.

39.0 COMPONENT SPECIFICATIONS

The Smart Meters shall be designed and manufactured using SMT (Surface Mount Technology) components, except for power supply components, LED / LCD etc., which are PTH type. All the material and electronic power components used in the manufacture of the Smart Meter shall be of highest quality and reputed makes so as to ensure higher reliability, longer life and sustained accuracy. The Components used for manufacture of Smart Meter shall be of high quality and the bidders shall confirm component specification as specified below in Annexure-III Bidders shall compulsorily fill Annexure-I, Annexure-II & Annexure-III for technical qualification.

Sl. no.	Component Function / Feature	Requirement	Make / origin
1	Current Element	E-beam /spot welded CT shall be provided in the phase element and in The neutral with proper isolation.	Any make or origin conforming to IS-2705.
2	Measurement /computing chips	The Measurement / computing chips used in the Smart Meter shall be with the Surface mount type along with the ASICs.	USA: Analog Devices, AMS, Cyrus Logic, Atmel, SAMES, Texas Instruments, Teridian; Japan: NEC, Freescale, Renesas; Holland: Phillps
3	Memory chips	The memory computing chips shall not be affected by the external parameters like sparking, high voltage spikes or electrostatic discharges.	USA: National Semi Conductor, Atmel, SAMES, Texas Instruments, Teridian, ST, Microchip; Japan: Hitachi,OKI,Freescale, Renesas; Holland / Korea:Phillps
4	Display modules	The display modules shall be well protected from the external UV radiations. The display shall be clearly visible over an angle of at least a cone of 70o.The construction of the modules shall be such that the displayed quantity shall not disturbed with the life of display. The display shall be TN type industrial grade with extended temperature range	Singapore: Bonafide Technologies; Korea: Advantek; Japan : Hitachi, SONY,Hijing,Truly Semiconductor; Chaina: Tianma
5	Communication modules	Communication modules shall be compatible for the RS 232 ports as per reference under 15959	USA: National Semi conductors, HP,ST, Texas Instruments, Agilent, Avago; USA / Korea: Fairchild; Holland/ Korea: Philips;Japan: Ligitek, Hitachi, Germany: Siemens,Tiwan: Everlight,

6	Optical port	Optical port shall be used to transfer the Smart Meter data to Smart Meter reading instrument. The mechanical construction of the port shall be such to facilitate the data transfer easily.	USA: National Semi conductors. Texas Instruments,HP, Agilent, Avago,Germany/USA:Osram;Japan: Hitachi, ,21; Germany:Siemens;Holland / Korea:Philips; Tiwan:Everlight,
7	Power Supply	The power supply shall be with the capabilities as per the relevant standards. Thepower supply unit of the Smart Meter shall not be affected in case the maximum voltage of the system appears to the terminals due to faults or due to wrong connections.	As specified.
8	Electronic components	The active & passive components shall be of the surface mount type & are to be handled &soldered by the state of art assembly processes.	USA: National Semi conductors, Atmel, Phillips, Texas Instruments, ST, Onsemi; Japan: Hitachi, Oki,Toshiba. Freescale; Korea: Samsung.
9	Mechanical parts	The internal electrical components shall be of electrolytic copper &shall be protected from corrosion, rust etc. The other mechanical components shall be protected from rust, corrosion etc. by Suitable plating / painting methods.	N.A.
10	Battery	Lithium-ion with guaranteed life of 10 years	Renata, Panasonic,Varta, , Sanyo, National,Tadiran, Sony,Duracell,Tekcell,Mitsubishi, EVE, SAFT , XENO
11	RTC / Micro controller	The accuracy of RTC shall be as per relevant IEC / IS standards	USA: Dallas, Atmel,Motorola, NEC,Teridian Renesas,-TexasInstruments, ST, Microchips,Epson; Holland / Korea:Philips;Japan: NEC, OKI,Hitachi, Mitsubishi,Freescale,

1. Tamper Logic and Display parameters will be liable to changes and will be informed subsequently during Observation.

2. Default programmable features should be available (like conversion / interchangeability between from Import Mode or Imp/Exp (net) Mode, or Prepaid Mode

3. GTP and Tamper logic will be confirmed after sample testing.

Tamper Logic-HT (Provisional)

Sl. No	Parameter	Occurrence	Restoration
1	**Power Related Tamper:**		
i)	**Power Failure**	If power goes off for more than persistence time	Power restores
	Logging Time	After 5 min	Immediate
2	**Voltage Related Tamper:**		
i)	**Invalid Voltage**	i) V3x > 70 % of Vref	i) V3x Ignored
		ii) Difference of angles between two phases ≤ 10 degrees	ii) Difference of angles between two phases > 10 degrees

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	Logging Time	After 5 min	After 5 min
ii)	**Missing Potential (Logging Phase Wise)**	i) Vx<15%of Vref	i) Vx>40% Vref
		ii)Current Ignored	ii) Current Ignored
	Logging Time	After 5 min	After 5 min
iii)	**Voltage Unbalance**	V3x>70% of Vref	V3x>70% of Vref
		(Vmax-Vmin)>10% of Vref	(Vmax-Vmin)<10% of Vref
	Logging Time	After 5 min	After 5 min
iv)	**High Voltage**	Any Vx>115%Vref	V3x<115%Vref
	Logging Time	After 5 min	After 5 min
3	**Current Related Tamper**		
i)	**CT Open (Logging Phase Wise)**	i) Iresidual>10% of Ib	i) Iresidual<10% of Ib
		ii) Phase current<2% of Ib	ii) Phase current ignored
		iii) Average Line Current ignored	iii) Average current>10%of Ib
	Logging Time	After 5 min	After 5 min
ii)	**CT Bypass**	I residual >10% Ibasic	I residual <10% Ibasic
		Ix>2% Ib	Phase current ignored
		Average line current: Ignored	Iavg> 10 % Ibasic
	Logging Time	After 5 min	After 5 min
iii)	**Over Current**	Any Ix > 150% of Ibasic	I3x < 150% of Ibasic
	Logging Time	After 5 min	After 5 min
iv)	**CT Reversal (Phase wise)**	Ix> 5%of Ib	Ix> 5% of Ib
		Direction: Negative	Direction: Positive
		Net Power factor > 0.5	Net Power factor > 0.5
	Logging Time	After 5 min	After 5 min
v)	**Current Unbalance**	Imax-Imin>30% of Imax for that period	Imax-Imin<29% of Imax for that period
		Average line current>5% of Ib	Average line current>5% of Ib
	Logging Time	After 5 min	After 5 min
4	**Other Tampers:**		
i)	**Low PF**	i) All Phase Currents > 10% of Ib	i) All Phase Currents > 10% of Ib
		ii) Average PF < 0.3	ii) Average PF > 0.3
	Logging Time	After 5 min	After 5 min
ii)	**Magnetic Tamper**	Whenever the Meter functionality gets affected on account of presence of any magnetic field, Meter shall log it as an Event and start recording at Imax if does not remain immune. Same shall be logged with date & time stamp.	
	Logging Time	30 sec (Approx)	30 sec (Approx)
iii)	**Cover Open**	On Removal of Meter Cover the Meter will lock Cover Open Event with Date and Time Stamp, it must be reflected in Auto Display mode	No restoration.

	Logging Time	Immediate	

In addition to above, Tamper Logic for 3 Phase 4 Wire Import/Net (Import - Export) Meter

	Neutral Disturbance and Magnet	Under any circumstances, if meter logs Neutral Disturbance or Magnetic Field Tamper event and starts recording at Imax then it will log in Import Register instead of Export Register.
	Current Reverse	If all three CT are in reverse, no Current Reversal Tamper should occur.
	Manual Resetting of Maximum Demand:	When Reset Button is pressed within an Integration Period, Rising Demand will not reset to Zero. The Demand will be registered for the entire Integration Period and will be logged as Current Max. Demand at the end of the Integration Period.

Please note:
V_{3x} Voltage in all Phases
V = Voltage In any Phase
I_{3x} = Current in all Phases
I_x= Current in any phase

Annexure-I
GTP for Meter

Sl.No.	Item Description	Manufacturer's Particulars
1.	Maker's name and country	To be specified by the Bidder
2.	Type of meter/model	To be specified by the Bidder
3.	Standard Applicable	IS 14697, IS 15959, CBIP 325,16444 Part II
4.	Accuracy/Interface class	0.5S
5.	Parameters displayed	As per Specification
6.	P.F. Range	Zero lag — unity — Zero Lead
7.	Basic Current (Ib) (.15A)	5A
8.	Maximum Current (Imax)	10A
9.	Minimum starting current	0.1% of I-basic
10.	Rated Voltage	110 V : Phase to Phase, 63.5 V : Phase to Neutral
11.	Meter Constant	To be specified by the Bidder
12.	Variation of voltage at which meter functions normally	70% to 120% of reference Voltage
13.	Rated Frequency	50 Hz±5%
14.	Power Loss in Voltage circuit (VA & watt) & Current circuits (VA)	Voltage Circuit :- Will not exceed 1.5W and 10VA per phase Current Circuit - Will not exceed 1.0 VA per phase
15.	Dynamic range	As per IS 14697/16444 Part-II

16.	MD reset Provisions	Possible to reset MD by any of the following options:-1 1.Remote MD reset 2.Manual MD reset 3.MD reset by Laptop 4.Auto Monthly Reset
17.	Display: a)Type of register b)No. of digit of display and height of character c)Auto display mode & scroll mode d)Type of push button for scroll mode	Display will be a)LCD b)7 digit 7 segment, height- 10x5mm c)As per approved sample d)Spring loaded push button
18.	Non volatile memory	To be provided as per Specification
19.	Details of provision for taking reading during power off condition	Through internal non rechargeable battery
20.	Principle of operation	As per technical Specification
21.	MD integration period	15 minutes
22.	Weight of meter	To be specified by the Bidder
23.	Dimensions	To be specified by the Bidder
24.	Warranty	5.5 years from the date of supply
25.	Outline drawings & leaflets	To be provided by the Bidder
26.	a)Remote meter- readout facility	Provision required
	b)Communication protocol used	DLMS
	c)Sealing provision for meter & optical port	To be provided as per Specification
	d)Baud rate of data transmission	9600 bps
	e)Required software to be resident in CMRI and BCS AND HES	To be provided by the Bidder
	f)Ultrasonic welding of body or any other technology which is equally or more efficacious	To be provided
	g)Manufacture Seal	To be provided
27.	Base Computer software	Compatible with windows 7 or above.
28.	Type test certificates	To be provided by the Bidder
29.	Time of day zones (selectable)	3 TOD Zones to be provided with a provision for 8 TOD Zones
30.	Whether meter measures both fundamental & harmonic energy	As per Specification
31.	Real time clock accuracy	Maximum drift ± 5 Minutes per annum.
32.	Battery for real time clock	It shall be Lithium-ion / Lithium battery having at least 10 years of life
33.	Anti tamper features	As per Tamper logic provided by WBSEDCL.

34.	Effect of accuracy under tamper conditions	As per technical specification
35.	Drift in accuracy of measurement with time	As per IS: 14697 & CBIP 325 /16444 (Part-II)
36.	Name plate details	As per specification
37.	Type of calibration	Software calibrated
38.	Type of mounting	Projection mounting
39.	Testing facility	Shall be available with manufacturer, details to be provided
.	Data retention by NVM without battery backup and un-powered condition	As per specification
41.	Type of material used:	
a.	Base	As per specification
b.	Cover	As per specification
c.	Terminal block	As per specification
d.	Terminal cover	As per specification
42.	Screw	
	i.Material	As per specification
	ii.Size	As per specification
43.	Internal diameter of terminal hole	5.5mm
44.	Centre to centre clearances between adjacent terminals	As per IS: 14697/16444 Part II
45.	Security profiles	
	a)Basic Security	To be provided
	b)Advance security	To be provided

Annexure-II

Pre-Qualification Conditions for Three Phase Static Meters

SL. No.	Particulars	Remarks
1	Bidders must have valid BIS certification for the offered meter.	Yes / No
2	Bidder preferably posses ISO 9001 certification	Yes / No
3	Bidder shall be manufacturers of static meters having supplied Static 1-ph or 3-phase meters with memory and LCD display to Electricity Boards / Utilities in the past 2 years.	Yes / No
4	Bidder has Type Test certificate for the Type of offered meter not more than 3 (three) years old	Yes / No
5	Bidders shall have dust free, static protected environment for manufacture, assembly and Testing.	Yes / No
6	Bidder shall have automatic computerized test bench for lot testing of meters.	Yes / No
7	Bidder has facilities of Oven for ageing test. .	Yes / No
8	Bidder shall submit certificate for immunity against magnetic influence of 0.2 T AC. & 0.5 T DC. from a NABL accredited Laboratory, for the same type of meter as offered.	Yes / No

Annexure III

SI No.	Component Function / Feature	As per Requirement
1	Current Element	
2	Measurement / Computing chips	
3	Memory chips	
4	Display modules	
5	Communication modules	
6	Optical port	
7	Power Supply	
8	Electronic components	
9	Mechanical parts	
10	Battery	
11	RTC / Micro controller	

ANNEXURE — IV

SI.	LIST OF DOCUMENTS TO BE SUBMITTED		
No.	DURING SAMPLE SUBMISSION		
1	Attested copy of type test reports from NABL accredited laboratory		
	Attested copy of BIS certificates of the same type of meter submitted as sample		
3	Attested certificates as regards material used for meter case, cover & terminal block.		
4	Annexure — II as per tender documents		
5	Annexure — III as per tender documents		
6	Operating manual of the meter submitted		

LT BULK Meter DISPLAY

12.3Auto Display Mode: In this mode, the below listed parameters shall be displayed in the following sequence:

26.1.1 LCD Test
26.1.2 Smart Meter Serial Number
26.1.3 Real Time & Date (DD/MM/YYYY)
26.1.4 Rising Apparent Demand with elapsed time while Active Import
26.1.5 Rising Apparent Demand with elapsed time while Active Export
26.1.6 No. of Power Failures
26.1.7 Cumulative Active Import Energy (Cumulative sign/legend must be given)
26.1.8 Cumulative Active Export Energy
26.1.9 Cumulative Reactive Energy – Quadrant-I
26.1.10 Cumulative Reactive Energy – Quadrant-II

26.1.11 Cumulative Reactive Energy – Quadrant-III
26.1.12 Cumulative Reactive Energy – Quadrant-IV
26.1.13 Cumulative Apparent Energy (Import)
26.1.14 Cumulative Apparent Energy (Export)
26.1.15 Apparent Maximum Demand (Import)
26.1.16 Apparent Maximum Demand (Export)
26.1.17 Inst. Voltages – Phase-wise (P-N)
26.1.18 Inst. Currents – Phase-wise
26.1.19 Inst Neutral Current
26.1.20 Signed Inst. Power Factor – Phase-wise
26.1.21 Inst. Net Power Factor
26.1.22 Inst. Apparent Power
26.1.23 Signed Active Power in KW
26.1.24 Signed Reactive Power in KVAR
26.1.25 Frequency
26.1.26 Cumulative Billing Count
26.1.27 Cumulative Tamper Count
26.1.28 Cumulative Power Off Hours since manufacturing
26.1.29 Power **OFF** Hours of present month
26.1.30 Phase Sequence & Phase Correspondences of Voltage & Current
26.1.31 Connection Check
26.1.32 Self Diagnosis
26.1.33 Cumulative Net Active Forward Import Energy
26.1.34 Cumulative Net Active Forward Export Energy

12.4 Push Button mode: Over & above the parameters of Auto Display Mode, the following parameters shall be displayed on pressing the push button. The Smart Meter display shall return to auto display mode (mentioned above) if the push button is not operated for approx. more than 6 seconds.

Import Mode

17.2.53 LCD Test
17.2.54 Smart Meter Serial Number
17.2.55 Real Time & Date (DD/MM/YYYY)
17.2.56 Rising Apparent Demand with elapsed time while Active Import
17.2.57 No. of Power Failures
~~17.2.58 Cumulative Active Import Energy (Cumulative sign legend must be given)~~
17.2.59 Cumulative Reactive Energy – Quadrant-I
17.2.60 Cumulative Reactive Energy – Quadrant-II
17.2.61 Cumulative Reactive Energy – Quadrant-III
17.2.62 Cumulative Reactive Energy – Quadrant-IV
17.2.63 Cumulative Apparent Forward Energy (Active Import)
17.2.64 Apparent Forward Maximum Demand (Active Import)
17.2.65 TOD wise Cumulative Active Forward Energy (Import)
17.2.66 TOD wise Cumulative Apparent Forward Energy (Import)
17.2.67 TOD wise Apparent MD (Import) with Date & Time
17.2.68 TOD wise Active MD (Import) with Date & Time

17.2.69 Inst. Voltages – Phase-wise (P-N)
17.2.70 Inst. Currents – Phase-wise
17.2.71 Inst Neutral Current
17.2.72 Signed Inst. Power Factor – Phase-wise
17.2.73 Inst. Net Power Factor
17.2.74 Inst. Apparent Power
17.2.75 Signed Active Power in KW
17.2.76 Signed Reactive Power in KVAR
17.2.77 Frequency
17.2.78 Cumulative Net Active Forward Import Energy
17.2.79 History 1 Cumulative Active Forward Import Energy
17.2.80 History 1 Cumulative Apparent Forward Import Energy
17.2.81 History 1 Apparent MD while Active Import with Date & Time
17.2.82 History 1 Average Power Factor
17.2.83 Cumulative Apparent MD (Import)
17.2.84 Last Billing Date & Time
17.2.85 Cumulative Billing Count
17.2.86 Cumulative Tamper Count
17.2.87 Cumulative Power Off Hours since manufacturing
17.2.88 Power **OFF** Hours of present month
17.2.89 Phase Sequence & Phase Correspondences of Voltage & Current
17.2.90 Connection Check
17.2.91 Self Diagnosis
17.2.92 Battery Status
17.2.93 Present Tamper Status(PT/CT/Other)
17.2.94 First Tamper Occurrence with Date & Time
17.2.95 Last Tamper Occurrence with Date & Time
17.2.96 Last Tamper Restoration with Date & Time
17.2.97 Cover Open Information with Date & Time
17.2.98 High Resolution Cumulative Forward Active Energy (Import) (3+4 in KWh)
17.2.99 High Resolution Cumulative Forward Reactive Energy – Quadrant-(Q1-Q3)(3+4 in KVArh)
17.2.100High Resolution Cumulative Forward Reactive Energy – Quadrant-(Q2-Q4)(3+4 in KVArh)
17.2.101High Resolution Cumulative Forward Apparent Energy (Import) (3+4 in KVAh)
17.2.102Present Voltage THD
17.2.103Present Current TDD
17.2.104Maximum Voltage THD with date & time
17.2.105Maximum Current TDD with date & time

Net (Import-Export)

12.3.44 LCD Test
12.3.45 Smart Meter Serial Number
12.3.46 Real Time & Date (DD/MM/YYYY)
12.3.47 Rising Apparent Demand with elapsed time while Active Import
12.3.48 Rising Apparent Demand with elapsed time while Active Export
12.3.49 No. of Power Failures
12.3.50 Cumulative Active Import Energy (Cumulative sign/legend must be given)
12.3.51 Cumulative Active Export Energy
12.3.52 Cumulative Reactive Energy – Quadrant-I
12.3.53 Cumulative Reactive Energy – Quadrant-II
12.3.54 Cumulative Reactive Energy – Quadrant-III

12.3.55 Cumulative Reactive Energy – Quadrant-IV
12.3.56 Cumulative Apparent Forward Energy (Active Import)
12.3.57 Cumulative Apparent Forward Energy (Active Export)
12.3.58 Apparent Forward Maximum Demand (Active Import)
12.3.59 Apparent Forward Maximum Demand (Active Export)
12.3.60 TOD wise Cumulative Active Forward Energy (Import)
12.3.61 TOD wise Cumulative Active Forward Energy (Export)
12.3.62 TOD wise Cumulative Apparent Forward Energy (Import)
12.3.63 TOD wise Cumulative Apparent Forward Energy (Export)
12.3.64 TOD wise Apparent MD (Import) with Date & Time
12.3.65 TOD wise Apparent MD (Export) with Date & Time
12.3.66 TOD wise Active MD (Import) with Date & Time
12.3.67 TOD wise Active MD (Export) with Date & Time
12.3.68 Inst. Voltages – Phase-wise (P-N)
12.3.69 Inst. Currents – Phase-wise
12.3.70 Inst Neutral Current
12.3.71 Signed Inst. Power Factor – Phase-wise
12.3.72 Inst. Net Power Factor
12.3.73 Inst. Apparent Power
12.3.74 Signed Active Power in KW
12.3.75 Signed Reactive Power in KVAR
12.3.76 Frequency
12.3.77 Cumulative Net Active Forward Import Energy
12.3.78 Cumulative Net Active Forward Export Energy
12.2.60 History 1 Cumulative Active Forward Import Energy
12.2.61 History 1 Cumulative Active Forward Export Energy
12.2.62 History 1 Cumulative Apparent Forward Import Energy
12.2.63 History 1 Cumulative Apparent Forward Export Energy
12.2.64 History 1 Apparent MD while Active Import with Date & Time
12.2.65 History 1 Apparent MD while Active Export with Date & Time
12.2.66 Cumulative Apparent MD (Import)
12.2.67 Cumulative Apparent MD (Export)
12.2.68 Last Billing Date & Time
12.2.69 History 1 Average Power Factor
12.2.70 Cumulative Billing Count
12.2.71 Cumulative Tamper Count
12.2.72 Cumulative Power Off Hours since manufacturing
12.2.73 Power **OFF** Hours of present month
12.2.74 Phase Sequence & Phase Correspondences of Voltage & Current
12.2.75 Connection Check
12.2.76 Self Diagnosis
12.2.77 Battery Status
12.2.78 Present Tamper Status(PT/CT/Other)
12.2.79 First Tamper Occurrence with Date & Time
12.2.80 Last Tamper Occurrence with Date & Time
12.2.81 Last Tamper Restoration with Date & Time
12.2.82 Cover Open Information with Date & Time
12.2.83 High Resolution Cumulative Forward Active Energy (Import) (3+4 in KWh)
12.2.84 High Resolution Cumulative Forward Active Energy (Export) (3+4 in KWh)
12.2.85 High Resolution Cumulative Forward Reactive Energy – Quadrant-I (3+4 in KVArh)
12.2.86 High Resolution Cumulative Forward Reactive Energy – Quadrant-II (3+4 in KVArh)
12.2.87 High Resolution Cumulative Forward Reactive Energy – Quadrant-III (3+4 in KVArh)
12.2.88 High Resolution Cumulative Forward Reactive Energy – Quadrant-IV (3+4 in KVArh)

12.2.89 High Resolution Cumulative Forward Apparent Energy (Import) (3+4 in KVAh)
12.2.90 High Resolution Cumulative Forward Apparent Energy (Export) (3+4 in KVAh)
12.2.91 Present Voltage THD
12.2.92 Present Current TDD
12.2.93 Maximum Voltage THD with date & time
12.2.94 Maximum Current TDD with date & time

17.3 TOD Wise

12.3.13 TOD wise Cumulative Active Forward Energy (Import)
12.3.14 TOD wise Cumulative Active Forward Energy (Export)
12.3.15 TOD wise Cumulative Apparent Forward Energy (Import)
12.3.16 TOD wise Cumulative Apparent Forward Energy (Export)
12.3.17 TOD wise Apparent MD (Import) with Date & Time
12.3.18 TOD wise Apparent MD (Export) with Date & Time
12.3.19 TOD wise History 1 Cumulative Active Forward Energy (Import)
12.3.20 TOD wise History 1 Cumulative Active Forward Energy (Export)
12.3.21 TOD wise History 1 Cumulative Apparent Forward Energy (Import)
12.3.22 TOD wise History 1 Cumulative Apparent Forward Energy (Export)
12.3.23 TOD wise History 1 Apparent MD (Import) with Date & Time
12.3.24 TOD wise History 1 Apparent MD (Export) with Date & Time

Power OFF Mode Display:

Smart Energy Meter Serial No.
Real Time
Date
Cumulative Active Import Energy
Cumulative Active Export Energy
History 1 TOD wise Active import energy
History 1 TOD wise Active export energy
History 1 TOD wise Apparent import Energy
History 1 TOD wise Apparent export Energy
History 1 TOD wise Maximum Demand in Apparent while Active import
History 1 TOD wise Maximum Demand in Apparent while Active export
Cumulative Billing Count
Cumulative Tamper Count
Cover Open Information with date and time

Display for Auto and manual mode must be listed by two headers:

"Auto Display Mode"
"Push Button Mode"

Display: Other requirements:

Each parameter shall be on Smart Energy Meter display for 10 sec and the time gap between two auto display cycles shall be 20 sec.

The register shall be able to record and display starting from zero, for a minimum of 1500 hours, the energy corresponding to rated maximum current at reference voltage and unity power factor. The register shall not roll over in between this duration.
No decimal is required for main kWh, kVAh, kVarh (lag & lead) display.
Push button mechanism shall be of high quality and shall provide trouble free service for a long span of time.
Up and Down scrolling facility of display parameters shall be there for Push Button Mode.

18.0 **For OBIS Codes of parameters, specified in this Technical Specification, other applicable standards** may be referred, in case those are not available in the standards referred in this document.

TAMPER LOGIC (Provisional) for **LT BULK Meter0**

Sl. No.	TAMPERS	Occurrence Condition	Restoration Conditions	Occurrence Time (Min/Sec)	Restoration Time (Min/Sec)
A	import Mode:				
1	Invalid Voltage	V_{3x}> 60% V_{ref} & <115% V_{ref}	V_{3x}> 60% V_{ref} & <115% V_{ref}	min.	5 min.
		Angle difference of any two phases <±10°	Angle difference of any two phases > ±100		
2	Missing Potential	Any V_x < 30% V_{ref} other phase voltage >40% V_{ref} & <115% V_{ref}	Any V_x > 40% V_{ref} other phase voltage >40% V_{ref} & <115% V_{ref}	5 min.	5 rain.
		Current >10% I_b	Current >10% I_b		
		Missing Potential detection will be Phase wise			
3	High Voltage	Any V_x > 115% V_{ref}	V_{3x} < 115% V_{ref}	5 min.	5 min.
		Current ignored	Current ignored		
4	Voltage Unbalance	V_{3x} > 70% Vfef & <115% Vref	V_{3x} > 70% V_{ref} & <115% V_{ref}	5 min.	5 min.
		(V_{max} — V_{min}) > 30% V_{ref}	{V_{mx}. — V_{min}} < 30% V_{ref}		
5	CT Open	$I_{Residual}$ >20% I_b	$I_{Residual}$ < 20% I_b	5 min.	5 min.
		I_X < 2% Ib.	I_x > 2% I_b		
		Average line Current :Ignored	Average Phase Current > 10%Ib		
		V_{3x} > 70% V_{ref} & <115% V_{ref}	V_{3x} > 70% V_{ref} & <115% V_{ref}		
		CT open detection will be phase wise			
6	CT Bypass	$I_{Residual}$ >20% Ib	$I_{Residual}$< 20% I_b	5 min.	5 min.
		I_{3x} > 2% I_b	Phase Current Ignored		
		Average line Current :Ignored	Average Current>10 % I_b		
		V_x 70% V_{ref} & <115% V_{ref}	V_{3x} > 70% V_{ref} & <115% V_{ref}		
7	Over current	Any I_x> 150% I_{max}	I_{3x} < 150% I_{max}	5 min.	5 min.
		V_{3x}> 70% Vref & <115% Vref	V_{3x} 70% Vref & <115% Vref		
	Low PF	I_{3x} >10% Ib	I_{3x} > 10% Ib	5 min.	5 min.
		Average PF <0.3	Average PF >0.3		

		V_{3x}> 70% Vref & <115% Vref	V_{3x} > 70% Vref & < 115% Vref		
	Neutral Disturbance	Frequency < 45 Hz or > 55 Hz	Frequency is between >=45Hz or <=55Hz	20 Secs - 40 Secs	20 Secs -40 Secs
10	Voltage THD	When Voltage THD > 5% in any phase	When Voltage THD < 5% in all phases	5 min	5 min
11	Current TDD	When Current TDD > 5% in any phase	When Current TDD < 5% in all phases	5 min	5 min
12	Magnet	Whenever the Smart Energy Meter functionality gets affected on account of presence of any magnetic field, Smart Energy Meter shall log it as an event and start recording at Imax if does not remain immune. In Tamper Snap Shot 'max must be shown (either occurrence or restoration), with Date and Time stamp. if Smart Energy Meter detects magnetic tamper in "Export" mode, the energy increment shall be made in Import mode as per Vref, Imax and UPF.		20 Secs	20 Secs
13	Cover Open	On removal of Smart Energy Meter cover the Smart Energy Meter will log cover open event along with date and time	No restoration	Immediate	
	Since this is LT CT operated meter Residual current should take the neutral current into consideration.				
B	Export Mode				
	Neutral Disturbance and Magnet	In Export Mode, Smart Energy Meter recording must not start at Imax. Under any circumstances, if Smart Energy Meter logs Neutral Disturbance or Magnetic Field Tamper event and starts recording at Imax then it will log in Import Register instead of Export Register.		OEM's Choice	
	Manual Resetting of Maximum Demand:	When Reset Button is pressed within an Integration Period, Rising Demand will not reset to Zero. The Demand will be registered for the entire Integration Period and will be logged as Current Max. Demand at the end of the Integration Period.			

Please Note:-

V_{3x} = Voltage in all Phases
V_x = Voltage In any Phase
I_{3x} = Current in all Phases
I_x = Current in any phase

Problems and avoid penalties that can be imposed by electrical utilities. More information about the levels of harmonic can be found on the IEEE website.

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